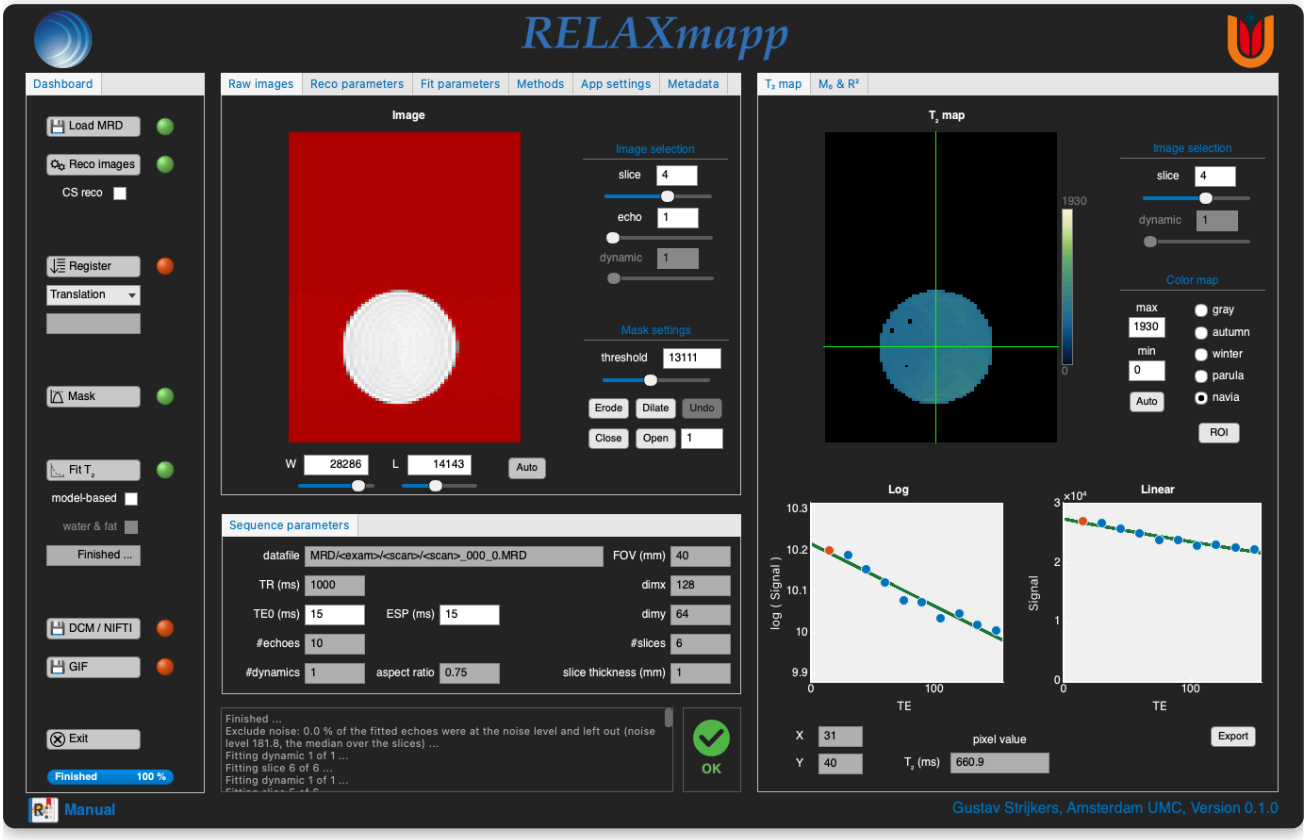
MAPapp — user manual

Version 0.1.0 (beta), October 2026



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1. Overview

MAPapp is an application with a graphical user interface to reconstruct relaxation maps from scans acquired with an MR Solutions preclinical MRI system. One window handles three kinds of scan:

- **T_1 scans**: inversion recovery, saturation recovery (a series of TRs) and variable flip angle series, fitted to T_1, M_0 and R^2 maps;
- **T_2 and T_2* scans**: multi-echo spin echo (T_2) and multi-gradient echo (T_2*), fitted to T_2 or T_2*, M_0 and R^2 maps; multi-gradient echo scans can also be separated into water and fat maps, with fat fraction and R_2* maps, and 3D multi-gradient echo scans give a magnetic susceptibility map (QSM);
- **ORACLE scans**: phase-cycled balanced steady-state free precession (bSSFP), which give T_1, T_2, M_0, off-resonance and R^2 maps from one scan.

The app recognizes the kind of scan when it is loaded and shows the map tabs and the Fit button that belong to it. The images are reconstructed, optionally registered, masked and fitted; the maps can be exported as DICOM, NIfTI, BIDS and GIF, with the values of a region of interest as a CSV file.

MAPapp is a stand-alone application: everything it uses, elastix and dcm2niix included, is inside it, and nothing else has to be installed.

2. Contact

MAPapp and this manual are written by:

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3. Disclaimer

This is a beta version. The software has been tested using mouse and rat data and phantoms acquired with an MR Solutions preclinical 7.0T system. However, this does not warrant that the functions contained in the program will meet your requirements or that the operation of the program will be uninterrupted or error-free. In case of questions or issues, please contact Gustav Strijkers.

4. Installation

macOS

Requirements: a Mac with Apple silicon (M1 or later), macOS 26 or later.

MAPapp comes as a disk image. Nothing else has to be installed: Python, elastix, dcm2niix and everything else the app uses are inside it.

1. Open the disk image and drag **MAPapp** onto the Applications folder in its window.
2. Open it from Applications. The app is not signed with an Apple Developer ID, so the first time macOS refuses it ("Apple could not verify ..."); click **Done**. Then open **System Settings > Privacy & Security**, scroll down to the line saying MAPapp was blocked, click **Open Anyway** and confirm. From then on it opens as any other app.

   Instead of step 2, the same in Terminal, once:

   ```
   xattr -dr com.apple.quarantine /Applications/MAPapp.app
   ```

The first start can take a little longer: macOS checks the app once.

Windows

Requirements: a PC with Windows 10 or 11, 64-bit.

MAPapp comes as an installer. Nothing else has to be installed.

1. Run the installer. It is not signed, so the first time Windows may stop it ("Windows protected your PC"): click **More info** and then **Run anyway**.
2. The installer puts the app in your user folder, without administrator rights (installing it for all users is an option), with a Start-menu entry and, if you choose, a desktop shortcut.

To remove the app, use **Settings > Apps**.

BART (optional)

With BART, the Berkeley Advanced Reconstruction Toolbox (mrirecon.github.io/bart), version 1.0.00 or newer, the model-based fits are available. Without BART, or with an older version, the model-based fit is switched off and compressed sensing runs on the built-in sigpy engine. The message box says so at start.

Installing BART on macOS with MacPorts

1. Install the Xcode command line tools:

   ```
   xcode-select --install
   ```

2. Install MacPorts from www.macports.org.
3. Install the packages:

   ```
   sudo port install fftw-3-single gcc14 libpng openblas flock gmake coreutils l
   ibblocksruntime
   ```

4. Download and unpack BART:

```
wget https://github.com/mrirecon/bart/archive/v1.0.00.tar.gz
tar xvfz v1.0.00.tar.gz
cd bart-1.0.00
```

5. Build and install:

```
export PATH=/opt/local/libexec/gnubin:$PATH
CC=gcc-mp-14 gmake
sudo gmake install
```

   The compiler name must match the gcc version installed in step 3: gcc14 gives gcc-mp-14, gcc15 gives gcc-mp-15.

6. Check that it works: `bart version`

Installing BART on macOS with Homebrew

1. Install the Xcode command line tools: `xcode-select --install`
2. Install Homebrew:

```
/bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/instal
l/HEAD/install.sh)"
```

3. Install the packages BART needs:

```
brew install gcc libpng fftw openblas make llvm libomp
```

4. Download and unpack BART as above (step 4 of MacPorts).
5. Build and install:

```
CC=gcc-15 MACPORTS=0 gmake
sudo gmake install
```

   The compiler name must match the version Homebrew installed; check it with `ls /opt/homebrew/bin/gcc-*`.

6. Check that it works: `bart version`

How MAPapp finds BART on macOS

`sudo make install` puts BART in `/usr/local/bin`, which is where the app looks first. It looks in these folders, in this order:

Folder	
`/usr/local/bin`	where BART's own `make install` puts it
`/usr/bin`	
`/opt/homebrew/bin`	Homebrew's
`/opt/local/bin`	MacPorts'

An app opened from the Finder or the Dock does not see the settings of your shell (such as `TOOLBOX_PATH` or `PATH` in your shell profile). If your `bart` is in another folder, a link to it in `/usr/local/bin` lets the app find it. In Terminal, once (Terminal asks for your password):

```
sudo mkdir -p /usr/local/bin
sudo ln -s /path/to/bart/bart /usr/local/bin/bart
```

Installing BART on Windows 11

On Windows, BART runs inside the Windows Subsystem for Linux (WSL); the app calls it as `wsl bart`.

1. Install WSL. Start a Windows PowerShell as administrator and run `wsl --install`. This installs WSL 2 and Ubuntu; a restart is needed. The first time you start the Linux prompt you are asked to create a user account.
2. Update the Linux distribution:

   ```
   sudo apt-get update
   sudo apt-get dist-upgrade
   ```

3. Install the packages BART needs (gcc 12 or newer):

   ```
   sudo apt-get install make gcc libfftw3-dev liblapacke-dev libpng-dev libopenb
   las-dev gfortran
   ```

4. Download, build and install BART:

   ```
   wget https://github.com/mrirecon/bart/archive/v1.0.00.tar.gz
   tar xvfz v1.0.00.tar.gz
   cd bart-1.0.00
   make
   sudo make install
   ```

5. Check from a Windows command prompt: `wsl bart version`

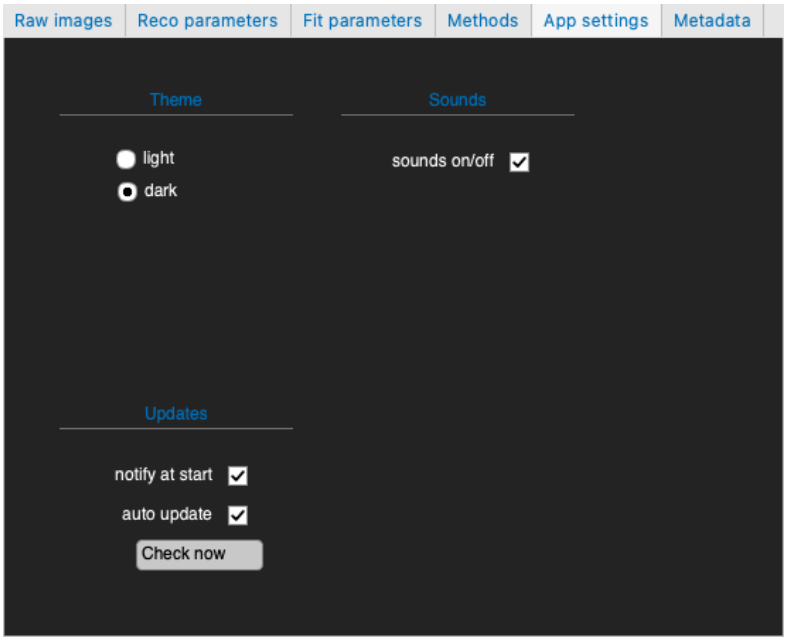
5. Running the software

Open **MAPapp** from the Applications folder (macOS), or from the Start menu or the desktop shortcut (Windows). The window appears within seconds. At its start the app looks for elastix and BART, and the message box reports what it found; "Ready" says it is done. Only **Load MRD** and **Exit** are on until a scan is loaded.

A scan given on the command line (an MRD file or its folder, as a program that starts MAPapp with a scan gives it) is loaded at once, as with **Load MRD**; a path the app cannot load gives a warning in the message box.

Updates, in the App settings tab (beside the Sequence parameters tab):

- **auto update**, ticked as it comes: new versions of MAPapp arrive by themselves. When one is ready, a window says so with what is new: **Install and Restart** installs it and starts the new MAPapp; **Later** installs it when you quit MAPapp, without asking again. The window does not come while a step runs, but when it has finished. Without an internet connection nothing is said and nothing waits.
- **Check now** looks for a newer version at once, also with **auto update** off; the message box says what it found. On the Mac, **Check for Updates...** in the app menu does the same.
- **notify at start**, ticked as it comes, counts only with **auto update** off: when a newer version is available, a window at the start offers it. **Download** gets the installer for this computer, **Skip this version** shows no window for that version again, **Later** asks again at the next start.



The App settings tab: theme, sounds and updates

Theme and sounds. The App settings tab, with its Theme, Sounds and Updates panels side by side, also chooses the window's theme, **light** or **dark**, and switches the sounds that go with a warning or an error (**sounds on/off**).

This manual. The **Manual** button at the bottom of the window opens this manual in the computer's PDF viewer.

The window has a fixed size. On a screen too small for it, the window fits the screen and can be scrolled.

Settings and log files

MAPapp keeps its files in a folder of its own, one per version, which it makes at its first start:

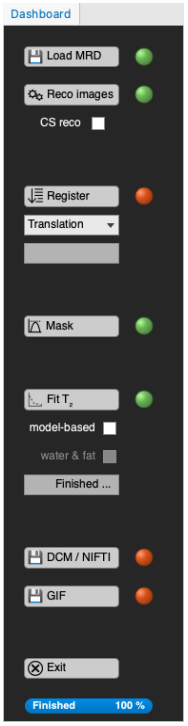
macOS	`~/Library/Application Support/MAPapp/<version>p`
Windows	`%APPDATA%\MAPapp\<version>p`

Settings. The settings are kept in `mapappSettings.json`. The file is written with the standard values the first time a version starts, with your choices from the version used before, and can then be edited by hand; changes take effect at the next start. It holds the version check, the folders where loading and the exports start (`mrdImportPath`, `defaultExportPath`), the theme, the sounds, the values the reconstruction, registration and fit fields start with, the color maps of the maps, and the number of log files kept. A value that cannot be used is replaced by its standard value, and the message box says so; a file that cannot be read at all is set aside as `mapappSettings-unreadable-<time>.json` and replaced by a new one.

Kept choices. MAPapp keeps these choices for the next start, and a new version takes them over: the theme, the sounds, auto update and notify at start, the color maps of the ten maps (T_1, T_2, M_0, R^2, water, fat, off-resonance, susceptibility, fat fraction, R_2^*), the registration method, min R-square, de-noise, de-ring, exclude noise, the ORACLE smoothing σ, the methods and parameters of the Methods 1 tab, the folders last used for loading and for the exports, and the lab values of the Metadata tab (step 9).

Log files. Every session writes a log file, `mapapp-log-<date and time>.txt`, to the same folder. The log keeps every message and the parameters of each reconstruction, registration and fit. Logs older than the newest 50 are removed at start.

6. Basic operation



The Dashboard

The MAPapp program operates from a single window with 5 panels.

Panel 1: Dashboard. The task buttons and the options that go with them. The tasks run from top to bottom; registration and masking can be left out. A green light shows a task done, yellow a task running, and red one not done or failed. At the bottom of the Dashboard one progress bar shows the progress of every task ("Registration 40 %", "Fit 40 %", "Finished"). The Fit button names what it fits: **Fit T_1** for a T_1 scan, **Fit T_2** for a T_2 or T_2* scan, **Fit $T_{1,2}$** for an ORACLE scan.

Panel 2: Raw images / Reco parameters / Fit parameters / Methods 1 / Methods 2. The images and the mask, the reconstruction and fit settings, and the water & fat and QSM methods (Methods 1). Methods 2 is kept for further methods.

Panel 3: Sequence parameters / App settings. The relevant acquisition parameters, and the app's settings (theme, sounds and updates). The count of the scan's points in the Sequence parameters tab is named after the kind of scan: **#echoes**, **#TIs**, **#TRs**, **#angles** (flip angles) or **#cycles** (phase cycles). For a T_2 or T_2* scan the first echo time (**TE0**) and the echo spacing (**ESP**) can be edited; changing them after a fit clears the maps.

Panel 4: the map tabs and Metadata. Only the map tabs of the loaded scan are shown; the **Metadata** tab (step 9) is always there, at the right end:

Scan	Map tabs
T_1 (inversion recovery, saturation recovery, variable flip angle)	T_1 map, M_0 & R^2
T_2 (multi-echo spin echo)	T_2 map, M_0 & R^2
T_2* (multi-gradient echo)	T_2* map, M_0 & R^2, Water & Fat when **water & fat** is ticked, FF & R2* when **water & fat** is ticked with a method other than B-spline, Susceptibility when **QSM** is ticked
ORACLE (phase-cycled bSSFP)	T_1 map, T_2 map, M_0 & R^2, Off-resonance

Panel 5: Messages. The program messages, newest on top, and the status: OK, BUSY, WARNING or ERROR.

The mouse scroll wheel steps through the slices: over the raw images those of the images, over the maps those of the maps. Over a slider the wheel moves that slider, one step per notch.

Step 1: Loading data

Press **Load MRD** to load the MRD raw data file. On the scanner computer (Windows) the first dialog of a session opens in the newest exam of the scanner's MRD folder, unless the settings name a folder (`mrdImportPath`); elsewhere it opens in the folder of the last file loaded.

MAPapp tells the kind of scan from the MRD parameters:

- **T_1**: an inversion recovery (Look-Locker FLASH), a saturation recovery (an array of TRs) or a variable flip angle series;
- **ORACLE**: a 2D or 3D balanced SSFP scan repeated with different RF phase increments (at least three);
- **T_2 or T_2***: a scan with more than one echo; a multi-gradient echo is fitted for T_2*, a multi-echo spin echo for T_2.

A scan that is none of these is refused, with the reason in the message box. Undersampled (compressed sensing) gradient echo scans are sorted with their trajectory table, the `.txt` file beside the MRD file. Retrospective's and P2ROUD's MRD exports load as well.

For multi-receiver coil data, select one of the MRD files; the app loads the others. The relevant acquisition parameters are shown in panel 3.



The Sequence parameters of a multi-echo spin echo scan

Echo times. For a T_2 or T_2* scan **TE0** and **ESP** are filled in when the scan is loaded. When the message box asks you to, check them before fitting.

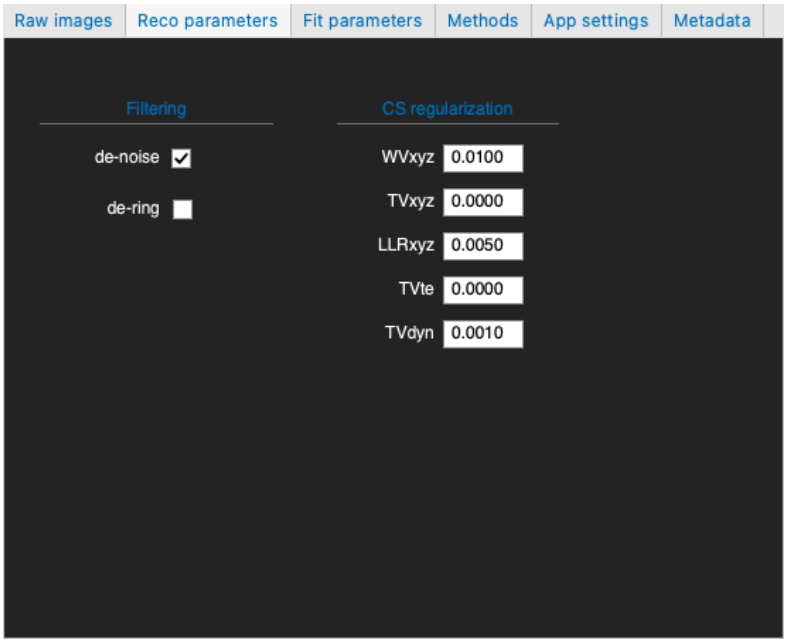
ORACLE scans. The message box says how many phase cycles the scan has, and the flip angle the scan was acquired with. Experiments at the start of the scan that only bring the signal to its steady state are left out. When the phase increments are not evenly spaced round the circle, a warning says so; the scan is fitted all the same.

Step 2: Image reconstruction

Loading reconstructs the images already. Press **Reco images** to reconstruct them again, for instance with compressed sensing. While a reconstruction runs, the button reads **Stop**; pressing it stops the reconstruction and leaves the earlier images and the parameters as they were.

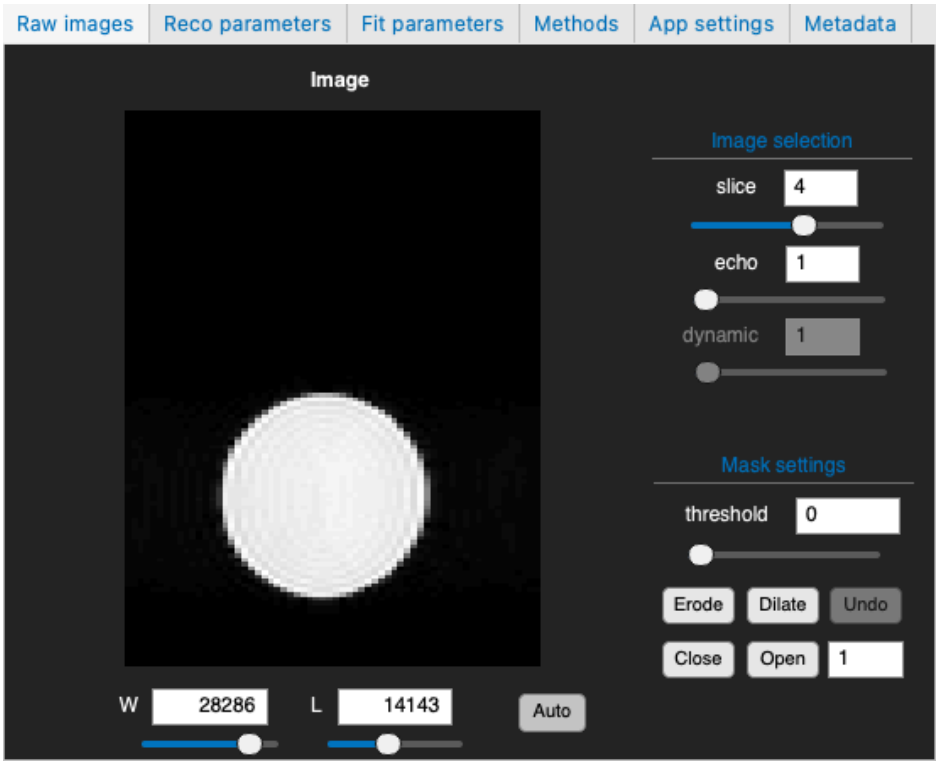
The standard reconstruction is a 2D or 3D FFT with PCA de-noising, on by default. When the **CS reco** checkbox on the Dashboard is checked, the images are reconstructed by compressed sensing with the weights of the Reco parameters tab: wavelets (**WVxyz**), total variation (**TVxyz**) and locally low rank (**LLRxyz**) in space, and total variation along the points of the scan (**TVte**: the echoes, inversion times, TRs or flip angles) and along the dynamics (**TVdyn**). For an ORACLE scan the two weights along the points and the dynamics are off: the profile over the phase cycles is what the fit reads. Compressed sensing runs with BART when it is installed, else with sigpy; the message box says which engine ran.

That tab also switches PCA de-noising (**de-noise**) and Gibbs ringing suppression (**de-ring**, off by default), which removes the ripple beside sharp edges.



The Reco parameters tab

The images can be inspected with the sliders of the Raw images tab: **slice**, the point of the scan (**echo** for a T_2 or T_2^* scan, **point** for the inversion times, TRs or flip angles of a T_1 scan, **cycle** for the phase cycles of an ORACLE scan) and **dynamic**. The window and level are set with the **W** and **L** fields and sliders, or with **Auto**.



The Raw images tab

Step 3: Image registration

NOTE: this step usually can be omitted.

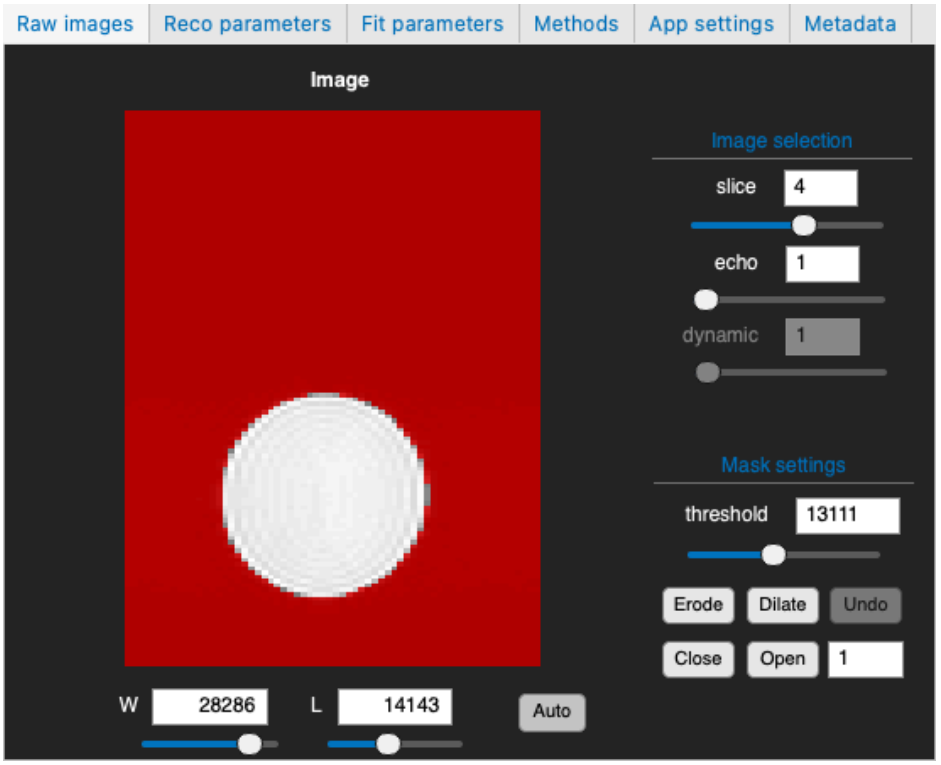
In case the images of the different points of the scan are not perfectly registered, press **Register** to register them as a **Translation**, **Rigid**, **Affine** or **B-Spline** transform (the list below the button). While the registration runs, the button reads **Stop**; pressing it stops the registration and leaves the images as they were. Registering clears the mask and the maps.

Step 4: Masking the background

To prevent unnecessary fitting of the background pixels, the images need to be masked first; without a mask every pixel is fitted, the background included.

Press **Mask** to mask them automatically. The red pixels indicate the regions that will be skipped during fitting. The threshold is found per slice for a 2D scan and once for the whole volume of a 3D scan (an ORACLE scan is masked on the image averaged over the phase cycles). The outermost layer of the object is left out, as are small specks of noise above the threshold; a dark region inside the object, such as air or a sinus, is skipped as well.

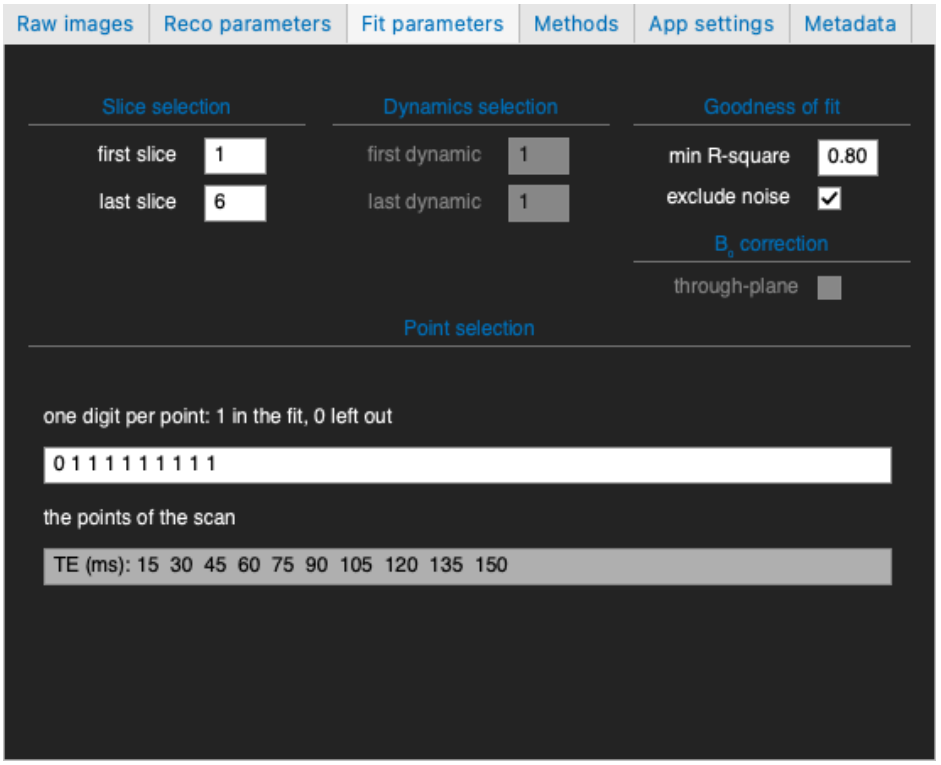
In case the automatic mask is not satisfying, the threshold can be set with the **threshold** field or slider; a threshold set by hand masks exactly the pixels above it, without the cleaning of the edge and the specks. For a 3D scan, a threshold typed or set with the slider applies to every slice; for a 2D scan it applies only to the slice on screen, and the other slices keep their automatic thresholds. Either way it applies to every dynamic. This holds for every kind of scan. The mask can be refined by **Erode**, **Dilate**, **Close** and **Open** operations, of the size in pixels in the field beside them; **Undo** takes the last one back. Any change to the mask after a fit, **Mask** included, withdraws the fitted maps: fit again to see and export them.



The mask: the red pixels are not fitted

Step 5: Fit parameters

The Fit parameters tab holds the fit settings.



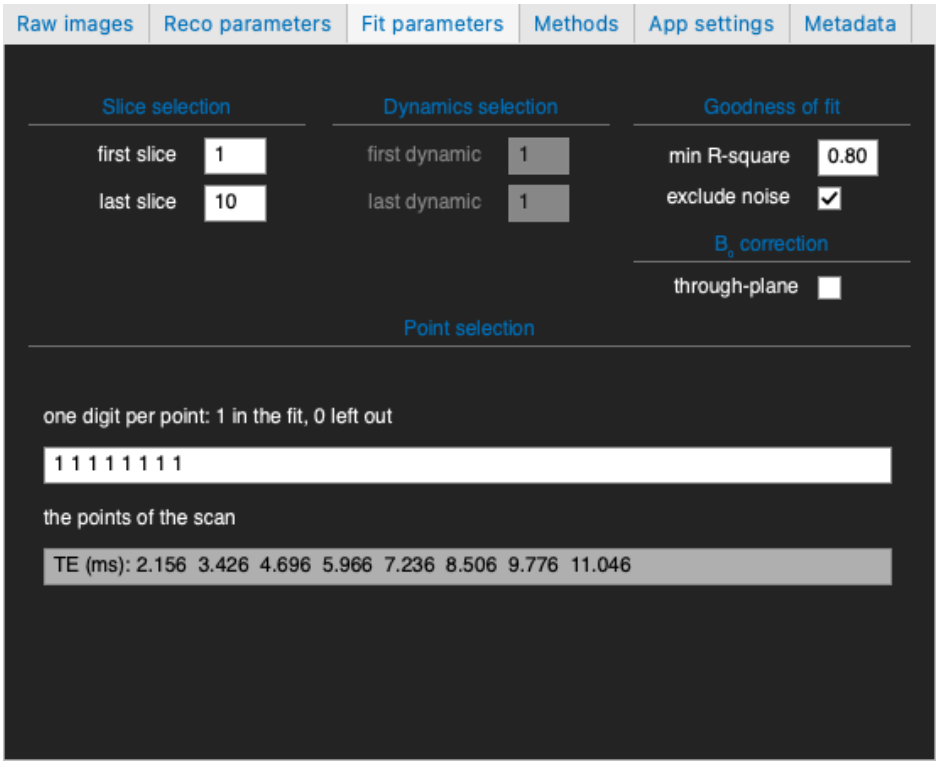
The Fit parameters tab of a multi-echo spin echo scan

Slice and dynamic selection. The first and last slice and dynamic that will be fitted (for a variable flip angle series, the cardiac frames take the place of the dynamics).

min R-square is the lowest R^2 accepted: where the fit of a pixel has a lower R^2 (0.80 by default), its values are set to zero in all maps.

Point selection. One digit per point of the scan, separated by spaces: **1** for a point in the fit, **0** for a point left out. `0 1 1 1 1 1 1 1` leaves the first of eight echoes out and fits the other seven. The line under the field lists the points in the same order: the echo times (TE), the inversion times (TI), the TRs, the flip angles or the phase increments. The field takes only 0s and 1s and is corrected when you press Return or leave it: other characters are removed, missing digits at the end become 1, digits beyond the scan's points are removed, and when fewer points are left in the fit than it needs (2 for a T_2 fit, 3 for a T_1 fit) all points are set back to 1. The message box says what was corrected. As the scan is loaded, every point is in the fit, except the first echo of a multi-echo spin echo. For an ORACLE scan the field is off: every phase cycle is fitted.

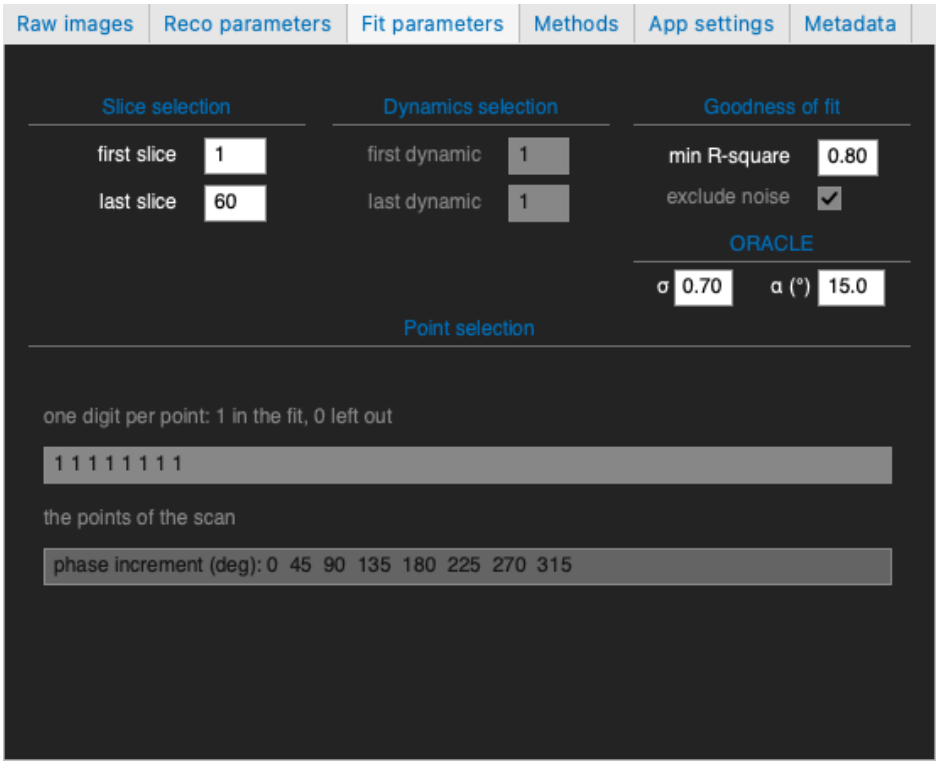

Raw images
Reco parameters
Fit parameters
Methods
App settings
Metadata
Slice selection
first slice 1
last slice 4
Dynamics selection
first dynamic 1
last dynamic 1
Goodness of fit
min R-square 0.80
exclude noise
B0 correction
through-plane
Point selection
one digit per point: 1 in the fit, 0 left out
1 1 1



The Fit parameters tab of a multi-gradient echo scan

ORACLE, for ORACLE scans, a panel in the place of the B_0 correction with two fields:

- **σ**, the smoothing in voxels, 0.7 by default, from 0 to 5. The T_1 and T_2 maps of a noisy scan are steadier when the fit smooths over the neighboring voxels inside the mask, over a width of σ voxels; 0 fits every voxel on its own. The off-resonance follows each voxel's own data. σ is kept for the next start.
- **α (°)**, the flip angle the fit uses, from 1 to 180°. It starts at 15° at every start of the app; the message box says the scan's own flip angle when the scan is loaded. T_1 depends strongly on the flip angle, so check it before fitting.



The Fit parameters tab of an ORACLE scan

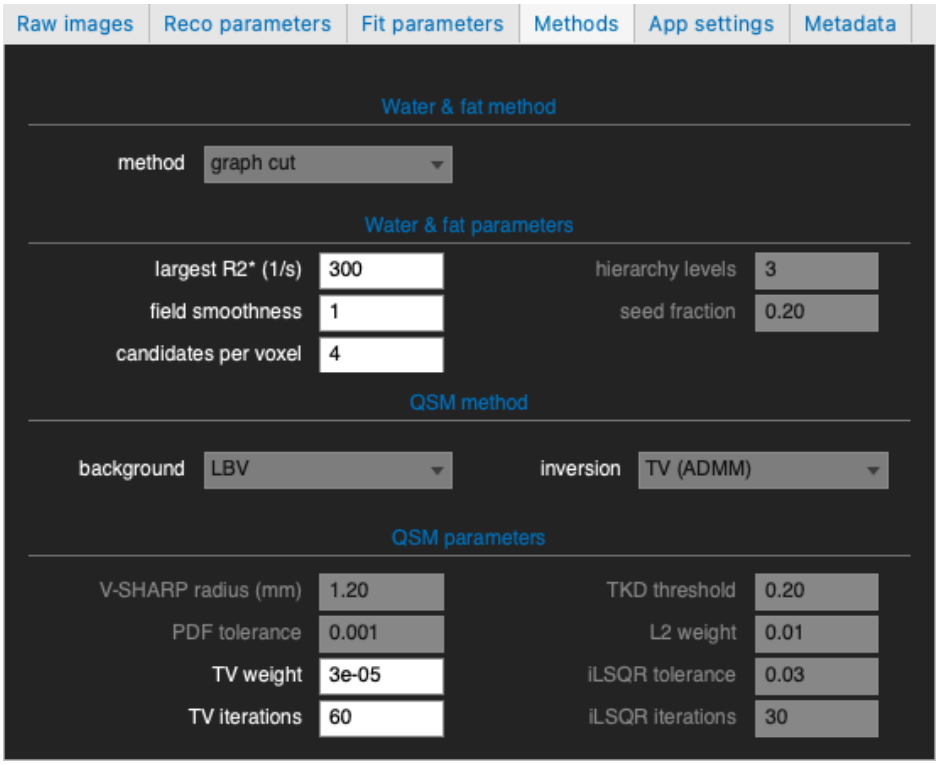
Water & fat, on the Dashboard: for a multi-gradient echo scan, tick **water & fat** before fitting to also separate water from fat. It needs at least 3 echoes, echo times at which fat and water can be told apart, and a scan without fat suppression. For a scan that does not allow it the box is off, and the message box says why when the scan is loaded. Echo times that tell fat from water only barely (an echo spacing close to a whole number of in-phase periods, section 8) leave the box on, with a warning in the message box at the load and at the fit: the maps can then be noisy and show patches where water and fat are swapped. The method is chosen in the Methods 1 tab (below).

QSM, on the Dashboard: for a multi-gradient echo scan the Dashboard shows a **QSM** tick in the place of **model-based**. Tick it before fitting to also make a map of the magnetic susceptibility χ (step 7). It is offered for 3D multi-gradient echo scans with at least 3 echoes whose files give the imaging frequency and the direction of the main field; for other scans the tick is off. The susceptibility map takes some tens of seconds on top of the T_2^* fit. An undersampled (compressed sensing) scan must be reconstructed with **CS reco** first. The QSM methods are chosen in the Methods 1 tab (below). For QSM, a scan of one average is the safer protocol; for a scan of several averages the message box says so.



The Dashboard of a 3D multi-gradient echo scan, with QSM ticked

The Methods 1 tab holds the methods of the water & fat separation and of QSM, with their parameters. Only the fields of the chosen methods are on; every choice and value is kept for the next start.



The Methods 1 tab

Water & fat method, in the upper half:

- **graph cut** (preferred, the standard choice): chooses the field map for the whole slice at once, weighing each pixel's fit against the smoothness of the field; it leaves the fewest patches where water and fat are swapped.
- **hierarchical IDEAL**: finds the field map from coarse blocks down to single pixels; the fastest, with a little more speckle.
- **region growing**: grows the field map from the most reliable pixel outwards to its neighbors.
- **B-spline**: a smooth field map over the slice, without R_2* and without the correction of the bipolar readout.

Graph cut, hierarchical IDEAL and region growing fit R_2* together with water and fat, and correct the phase difference between the odd and even echoes of a bipolar readout. Their parameters: **largest R2* (1/s)**, the highest decay rate searched (300 by default; 0 leaves R_2* out); **field smoothness**, how strongly neighboring pixels are held to similar fields (1 by default); **candidates per voxel**, the number of possible fields each pixel offers (graph cut, 4); **hierarchy levels**, the number of block sizes (hierarchical IDEAL, 3); **seed fraction**, the share of the brightest pixels the starting pixel is chosen from (region growing, 0.2).

QSM method, in the lower half: the **background** field removal (**LBV**, the standard choice, which keeps the tissue up to the mask's edge; **V-SHARP**; **PDF**) and the **inversion** that turns the field into χ (**TV (ADMM)**, the standard choice, with the least streaking; **TKD**, a quick preview; **closed-form L2**; **LSQR**). Their parameters: **V-SHARP radius (mm)**, **PDF tolerance**, **TV weight** and **TV iterations**, **TKD threshold**, **L2 weight**, and **iLSQR tolerance** and **iLSQR iterations**.

model-based, on the Dashboard (T_1 and T_2 scans): tick it to fit a model-based reconstruction with BART, the maps then being fitted to the k-space directly. It is offered when BART is available, for multi-echo spin echo scans and for 2D inversion recovery scans (not for saturation recovery, flip angle or 3D T_1 scans); an inversion recovery whose images were registered is fitted pixel-wise. It gives no R^2 map and does not apply min R-square.

Step 6: Fitting the maps

Press the Fit button (**Fit T_1**, **Fit T_2** or **Fit $T_{1,2}$**) to start the fit. While the fit runs, the button reads **Stop**; pressing it stops the fit and leaves the maps as they were before it. The maps fill in slice by slice while the fit runs. After the fit the maps show the middle slice.

A press that would give the same maps again, with nothing changed since the last fit, is skipped with a message in the message box; the same holds for **Reco images** and **Register**. When only an extra is new, for example **QSM** ticked after a fit, only that extra is calculated and the maps are kept. With **water & fat** ticked and a method that fits R_2* (all but the B-spline method), the T_2* map comes from the water & fat fit (section 7) and the single-exponential fit is left out.

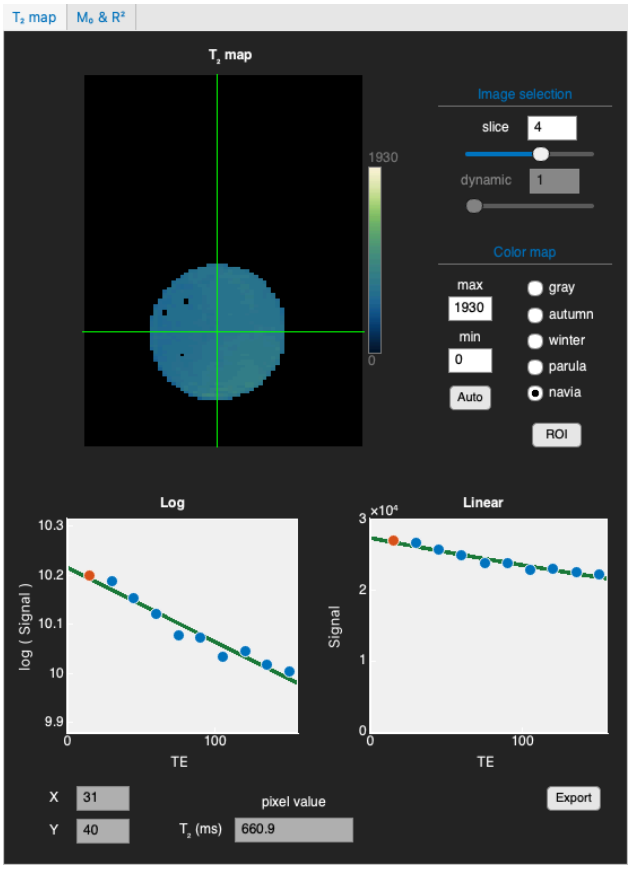
What each fit calculates is described in section 7. For every kind of scan, pixels outside the mask, pixels whose R^2 is below min R-square and pixels with a value outside the possible range (T_1 above 10 s, T_2 above 5 s) are zero in all maps.

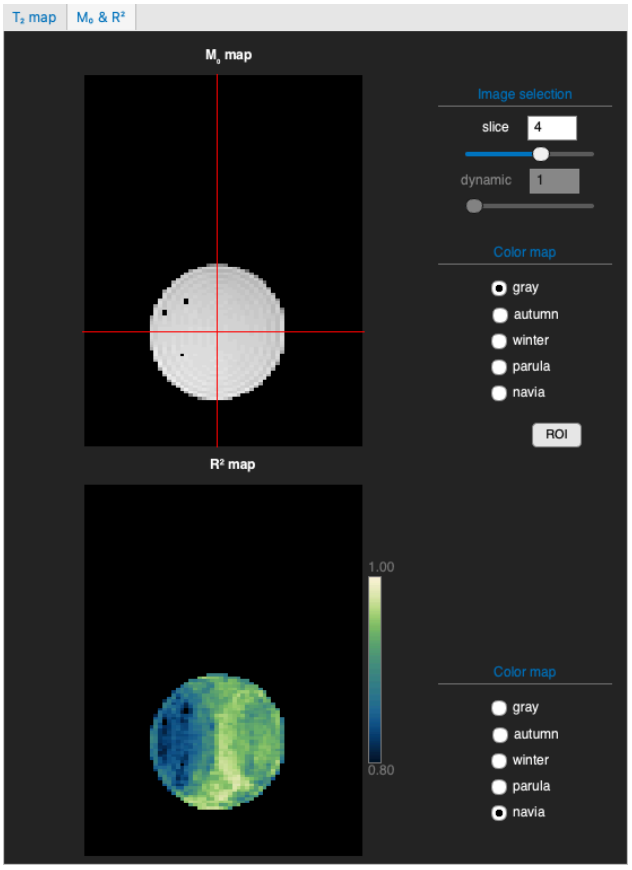
Step 7: The map tabs

Each map tab shows a map on a scale set with **max** and **min**, or with **Auto**, and a color map chosen beside it (gray, autumn, winter, parula and navia; for the T_1 map lipari in the place of navia). The color bars stand beside the maps with their end values above and below them. The slice and dynamic sliders of the map tabs move together: every tab shows the same slice and dynamic.

Auto sets the scale from the values of the slice shown: the T_2 map from 0 to twice its mean T_2; the T_1 map of a T_1 scan from 0 to twice the median T_1; the T_1 map of an ORACLE scan from 0 to twice its mean T_1; the off-resonance map over its whole range (below).

T_2 map | **M_0 & R^2**

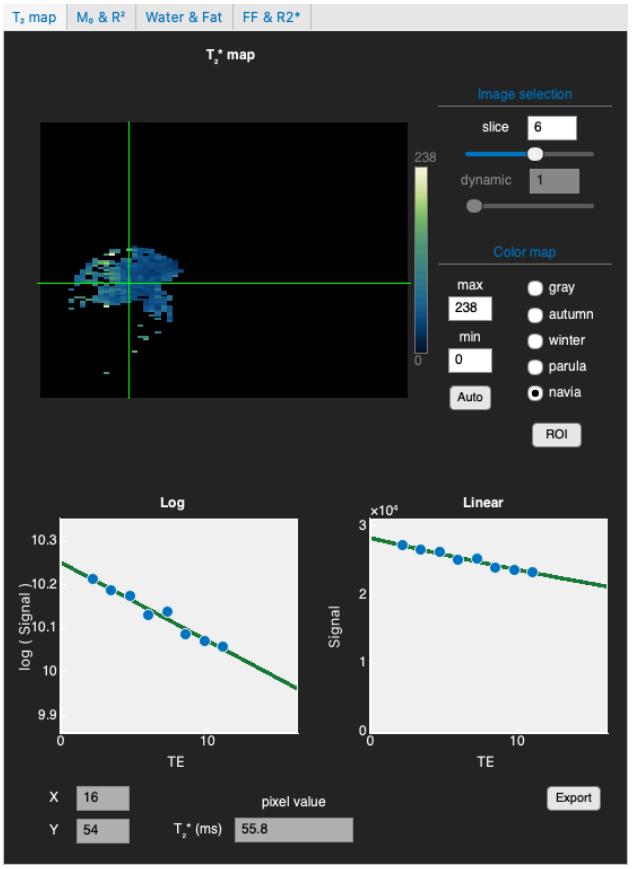




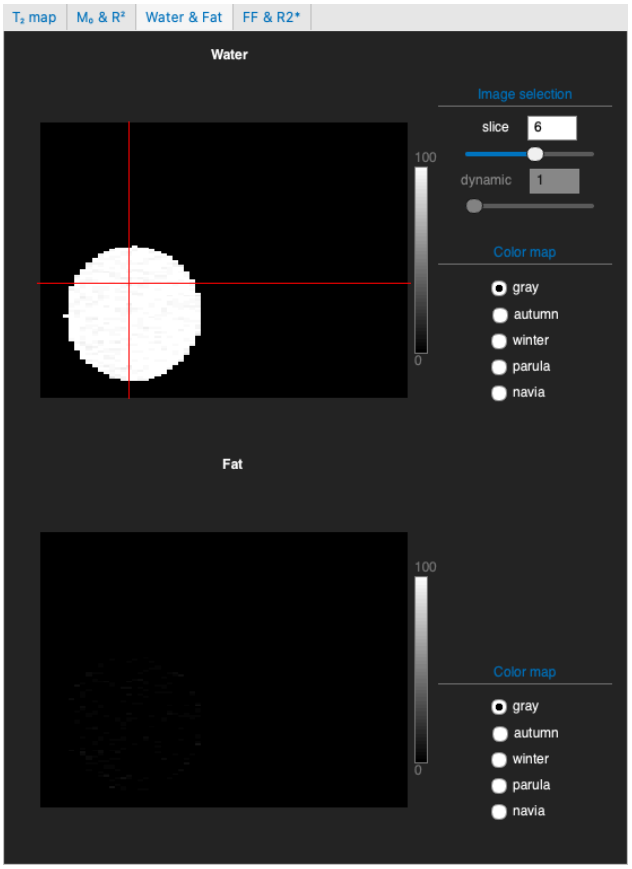
M_0 & R^2: the M_0 map (the signal the fit gives at time zero, or the profile's scale for ORACLE) and the R^2 goodness-of-fit map, its color bar from the R^2 cut-off to 1.

T_2* and water & fat. Multi-gradient echo scans are fitted in the same way as spin echo scans; the maps, the fields and the plots then say T_2*. With **water & fat** ticked, the Water & Fat tab shows the water and fat maps in percent of the total signal, each with its own color map and a color bar from 0 to 100 %.

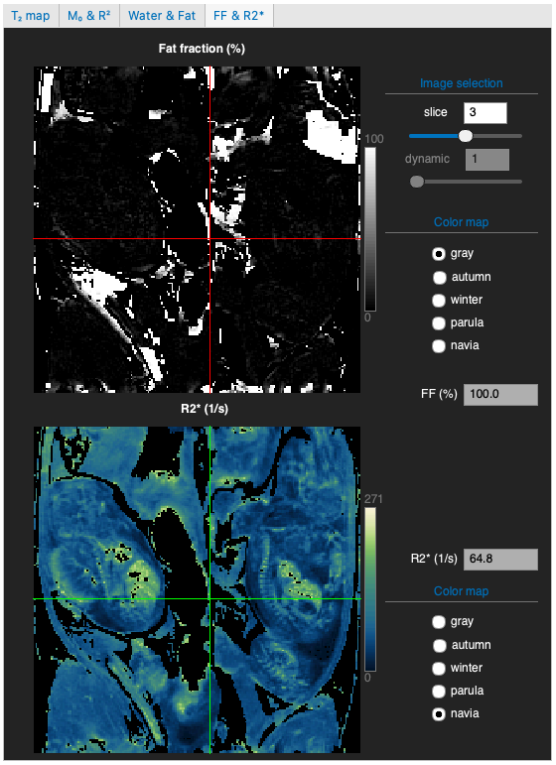
T_2* map



Water & Fat

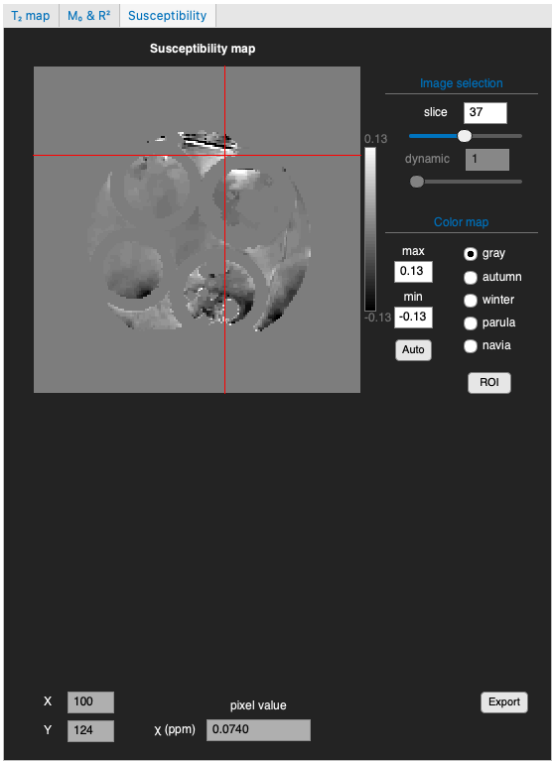


FF & R2* (with **water & fat** ticked and graph cut, hierarchical IDEAL or region growing chosen in the Methods 1 tab): the fat fraction map in percent, from 0 to 100 % (gray by default), above the R_2* map in 1/s (navia by default), whose color bar runs from 0 to the largest R_2* of the map. Each map has its own color map, kept for the next start. The tab has no **max**, **min** or **Auto**; its slice and dynamic sliders move with those of the other map tabs. Extreme R_2* pixels are set to 0, and the message box says how many. A click on either map places the cross-hair (or adds to the ROI) on both; the fields **FF (%)** and **R2* (1/s)** then show the fat fraction and R_2* of the pixel, or the average over the ROI, inside the mask.



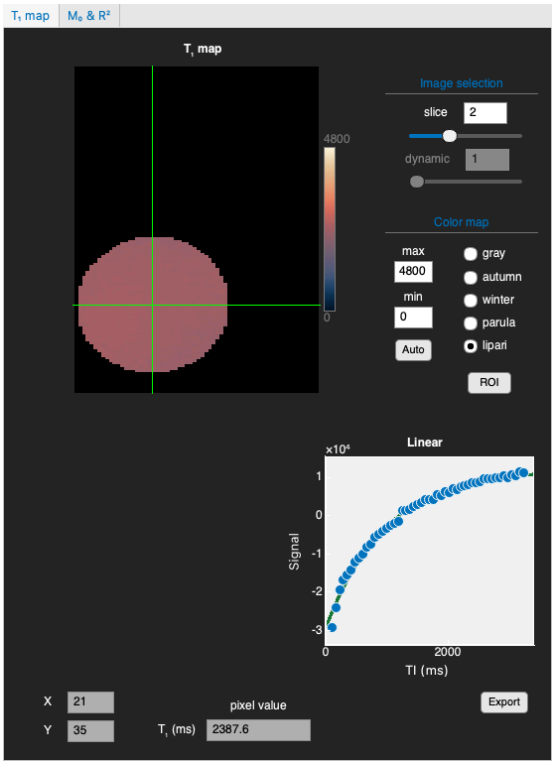
The fat fraction and R_2* maps of a multi-gradient echo scan of a mouse abdomen

Susceptibility (with **QSM** ticked): the map of the magnetic susceptibility χ in ppm, relative to the mean over the mask (the mean is zero), gray by default. Paramagnetic regions (iron, blood) are positive, diamagnetic regions (calcium, myelin) negative. Its **Auto** scale is symmetric about zero, up to the largest value of the map. Voxels where the scan's phase cannot be used are zero, and so are extreme pixels; the message box says how many. The pixel value is χ in ppm; there is no signal plot for this map.



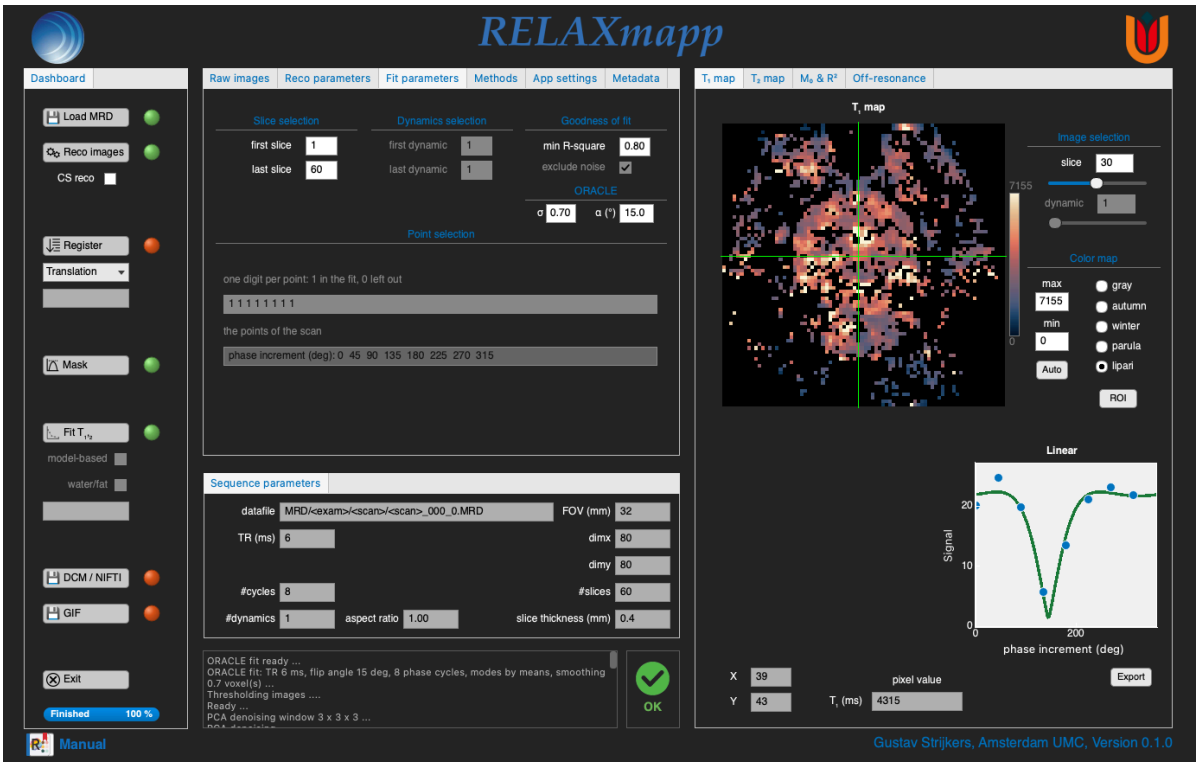
The susceptibility map of a 3D multi-gradient echo scan of a tube phantom

T_1 map of a T_1 scan: the plot below the map shows the signal along the inversion times, TRs or flip angles. The signal of an inversion recovery is shown with the sign the fit restored.



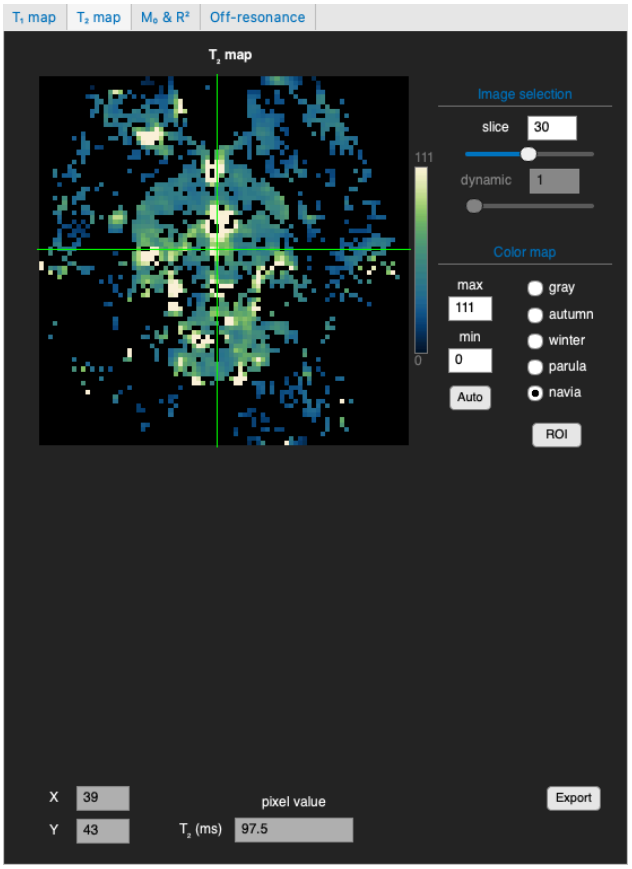
The T_1 map of an inversion recovery scan of a phantom, with the signal of a pixel along TI

ORACLE scans have four map tabs: **T_1 map**, **T_2 map**, **M_0 & R^2** and **Off-resonance**. The off-resonance map is in Hz, from −1/(2 TR) to +1/(2 TR): an off-resonance beyond that range shows up as one within it. Its Auto scale is that whole range.

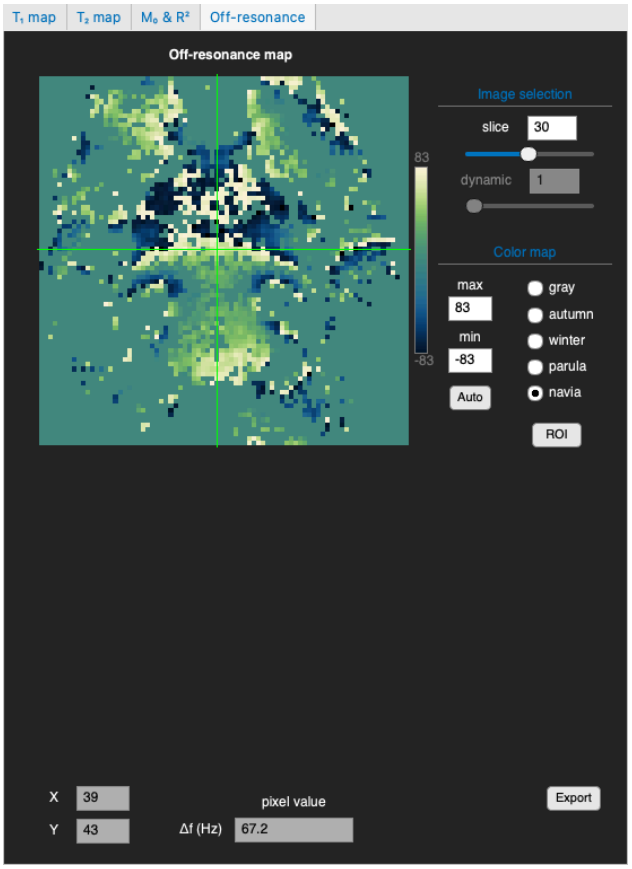


The window after fitting an ORACLE scan of a rat brain

T_2 map



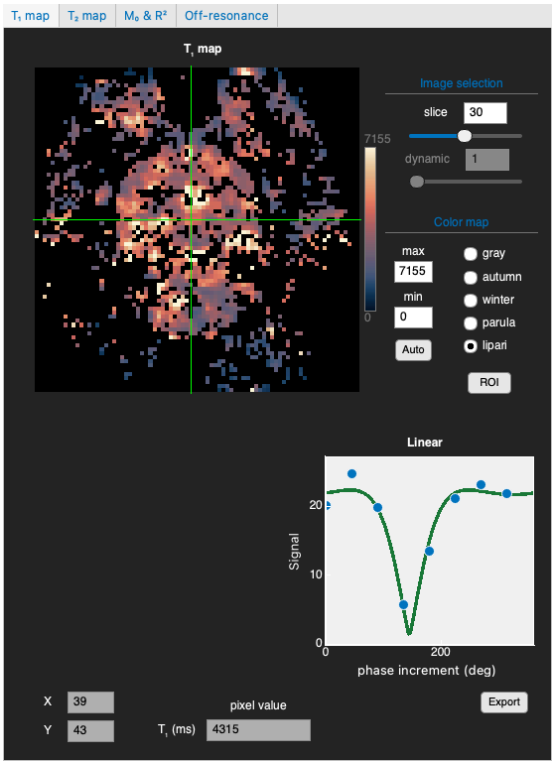
Off-resonance



Step 8: Values of pixels and ROIs

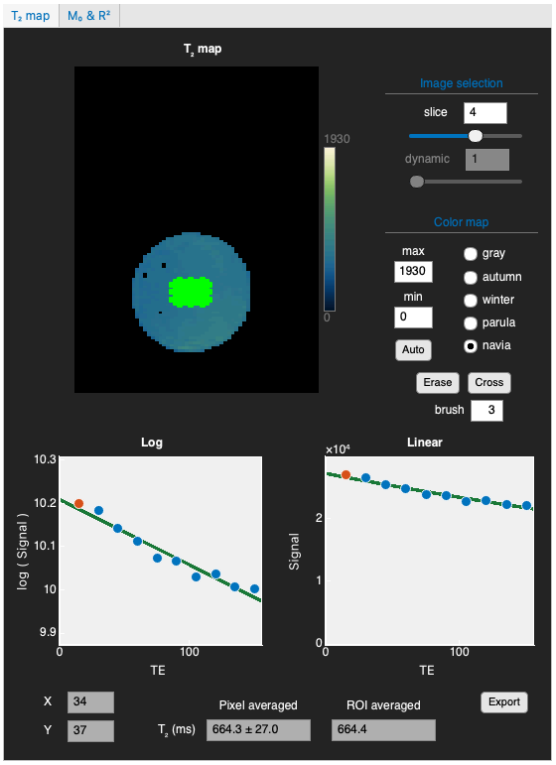
When you click on a map a cross-hair is shown, with the values of that pixel (**pixel value**) and a plot of its signal: symbols for the measured points, a line for the fitted curve. A click outside the mask shows nothing.

- **T_2 and T_2* scans**: log(signal) and signal as a function of TE. A red symbol shows an echo left out of the fit: by the point selection (step 5), because its signal was at the noise level (**exclude noise**), or because the B_0 correction left too little of it.
- **T_1 scans**: the signal along TI, TR or flip angle; a red symbol shows a point left out by the point selection.
- **ORACLE scans**: the T_1 map tab plots the size of the signal over the phase increment, 0 to 360°, with the profile that the fitted T_1, T_2, M_0 and off-resonance predict as the line. The T_1, T_2 and off-resonance of the pixel are shown in the T_1 map, T_2 map and Off-resonance tabs.



An ORACLE pixel: its profile over the phase increments, with the fitted profile

You can toggle between the cross-hair and region-of-interest (ROI) values with the **ROI** / **Cross** button. A single ROI can be 'painted' on the map, in one or more slices, by mouse clicks on the image, with the brush size in pixels set below the button.



An ROI painted in a phantom, with its signal and the fitted curve

For the ROI, two values are calculated:

- **Pixel averaged**: the average of the values of the pixels; for T_1 and T_2 scans this is 1 / mean(1/T) over the pixels, with its spread; for ORACLE scans and for χ the mean of the map's values.
- **ROI averaged**: the model is fitted to the average signal of the ROI, producing a single value for the ROI.

The ROI can be removed with **Erase**. **Export** writes the ROI-averaged signal to a CSV file, `roiData-<tag>.csv`, in a folder you choose: the echo times, inversion times, TRs, flip angles or phase increments, beside the signal.

Step 9: Metadata

The **Metadata** tab, the rightmost of the map tabs (always shown, whichever map tabs the scan has), holds the record of the scan, which the DCM / NIFTI and GIF exports write with the maps. Its fields stand in three columns:

- **Subject:** animal ID, species, strain and its RRID, sex, age, weight, group and time point;
- **Session:** project, ethics license, date and time, operator, institution, scanner, anesthesia, heart and respiration rate, and temperature, with a field for notes;
- **Contrast:** agent, concentration, dose, volume, route and the dynamic it was injected at.



The Metadata tab, as a load filled it

At a load the app fills in what the scan's files say: the animal ID and the time point (from the patient name typed at the console: `20240212_926752_D0` is animal 926752, day 0), the date and time, the institution and the scanner, and the project. What the console leaves at its standard values (sex O, weight 0, the login as the operator) is left empty.

What was filled in for an earlier scan of the same exam is filled in again: the animal's details are typed only once per exam. The date, the heart and respiration rate, the temperature, the notes and the injection's dynamic belong to one scan and are never taken from another.

Your institution, operator, anesthesia, species, strain and strain RRID are kept for later exams: they show, in blue, from the start, and a scan's own value takes their place where it has one.

A value the app filled in is shown in blue; every field can be corrected, and then counts as yours. Species, sex and route are lists to choose from, or to type in; numbers are kept with their units (a number field holding something else is edged in red). The species and sex are written with their codes (NCBI Taxonomy, PATO), so the record can be read by other software as well as by people.

Step 10: Exporting the maps

There are two ways to export the maps, and one for the ROI data.

1. **DCM / NIFTI**: exports the maps in DICOM and NIfTI format for further processing in third-party software, and the relaxation maps also as BIDS quantitative maps, in seconds (the DICOM and NIfTI maps are in ms). The DICOM files carry the species, strain, sex, age and weight of the animal where the Metadata tab has them.
2. **GIF**: exports the maps as animated GIF files, one frame per slice, or one GIF per slice with a frame per dynamic, in the color maps and scales shown, with two pictures of the app's window.
3. **ROI data**: the **Export** button of the map tab (step 8).

The maps each kind of scan exports:

Scan	DICOM and NIfTI	BIDS	Folder name
T_1	T_1, M_0, R^2	`_T1map`	`<tag>T1`
T_2	T_2, M_0, R^2	`_T2map`	`<tag>T2`
T_2^*	T_2^*, M_0, R^2; with water & fat: water, fat, and with graph cut, hierarchical IDEAL or region growing also the fat fraction (%) and R_2^* (1/s); with QSM: susceptibility χ (ppm)	`_T2starmap`; `_FFmap` (%) and `_R2starmap` with the fat fraction and R_2^*; `_Chimap` with QSM	`<tag>T2`
ORACLE	T_1, T_2, M_0, R^2, off-resonance (Hz)	`_T1map`, `_T2map`	`<tag>ORACLE`

The GIF export writes the maps the window shows as animations, the fat fraction and R_2^* included. BIDS has no name of its own for a fat fraction map; the app names it `FFmap`, as its DICOM and NIfTI series, in percent.

When the MRD file lies in the scanner's folders, …`/MRD/<exam>/<scan>`, the exports go to …`/Data/<exam>/DICOM`, `Nifti`, `BIDS` and `GIF`, each into a new numbered folder; otherwise the app asks for a folder. The folder dialog opens in the folder chosen last. The folder and file names the app makes from the scan's name hold no spaces: a space, or any character other than a letter, a digit, `-`, `_` or `.`, becomes `_`.

The DCM / NIFTI and GIF exports also write, in …`/Data/<exam>/Metadata/<tag><kind>/<n>` (or under the folder asked for), a new numbered folder each time:

- `metadata.json`: the Metadata tab's record;
- the report of the export, as `.json` and `.txt`: the app and its libraries, the scan and a checksum of each of its files, its acquisition, every step of the session with its parameters, the reconstruction, registration and fit settings, the statistics of each map per slice and dynamic within the mask, the points used, for the GIF export the scales and color maps, and every file the export wrote with its checksum;
- `ro-crate-metadata.json`: the same as an RO-Crate (www.researchobject.org/ro-crate), a standard description of the folder for data repositories.

The paths in them are relative to the exam folder (or the folder asked for), so they hold no names of your own folders. The ROI data export writes no record.

Step 11: Exit

Press **Exit** to shut down the program. Closing the window does the same, also while a step runs: the reconstruction, registration or fit is stopped, the window disappears, and the program closes as soon as the step has ended.

7. What is calculated

T_2 and T_2*. Each pixel is fitted to

> *Signal = M_0 exp(–TE / T_2)*

For multi-gradient echo data the same equation gives T_2*. With the B_0 correction the equation also holds the signal lost through the slice. R^2 compares the fitted curve with the measured signal. With water & fat ticked and a method that fits R_2* (all but the B-spline method), the T_2* map is 1000 / R_2* of the water & fat fit instead, with its M_0 and R^2: one decay shared by water and fat, so that the beating of fat against water is not taken for decay.

Water & fat. The complex echoes of a multi-gradient echo scan are separated into a water and a fat signal, fat with its spectrum of several peaks, together with the field map and (except with the B-spline method) the decay rate R_2*, after the phase difference between the odd and even echoes of a bipolar readout is corrected. The maps give water and fat each in percent of the total; with graph cut, hierarchical IDEAL and region growing also the fat fraction, fat / (water + fat) in percent, and R_2* in 1/s.

Properties. Every pixel holds water and fat that share one field and one R_2*. Because R_2* is part of the model, the fat fraction does not depend on the echo times, and the several fat peaks make it accurate at low and high fractions. The fat fraction comes from the signal magnitudes, which avoids phase errors; the field map comes first, from the complex echoes.

Limitations.

- **Water–fat swaps.** Water and fat differ in frequency by about 3.4 ppm, and a field offset of the same size looks the same in a single pixel. The methods choose the field by its smoothness across the slice; where that fails (large field steps at air or bone, tissue islands apart from the rest), water and fat can swap in a patch. Graph cut swaps least; hierarchical IDEAL is fastest and shows more single-pixel swaps; region growing depends on where it starts; B-spline assumes a smooth field and swaps most at field steps.
- **One R_2* for water and fat,** and a fixed fat spectrum (that of body fat). Tissue whose fat differs, or water and fat that decay differently, give small errors in the fat fraction.
- **T_1 weighting.** Fat recovers faster than water; a large flip angle or a short TR raises the fat fraction (section 8).
- **Noise.** Where the signal is low, the fat fraction is noisy, and very low fractions read a little high.
- **Echo times.** Echoes spaced at a whole number of in-phase periods, or too few of them, cannot separate water from fat (MAPapp then does not offer it); echoes spaced close to it separate them poorly (MAPapp warns), with noisy maps and swapped patches, and R_2* from them is no better than the T_2* map.
- **Readout.** Fat lies shifted along the readout by the chemical shift (about one pixel with the bandwidths of section 8); a bipolar readout's odd and even echoes are corrected, but a strong difference between them can remain at the edges of the object.
- **2D scans** are separated slice by slice; a swap can differ between neighboring slices.
- The B-spline method has no R_2* and no correction of a bipolar readout: its water and fat are weighted by the decay over the echoes.

Susceptibility (QSM). The phase of the echoes gives the magnetic field in every voxel. The part of the field that comes from outside the object is removed, and the remaining local field is converted into the susceptibility that causes it. χ is given in ppm, relative to its mean over the mask.

Properties. Each voxel's χ is the source of a small dipole field; the measured local field is the sum of the fields of all voxels. Iron and blood (paramagnetic) are positive, myelin and calcium (diamagnetic) negative. The map is relative: only differences between tissues are meaningful.

Limitations.

- **The inversion is ill-posed.** Some spatial patterns of χ make no field at all, so the conversion needs a constraint. TV (ADMM) assumes regions of constant χ and gives clean maps with sharp edges, but it flattens fine texture and lowers small, strong sources; TKD is quick but shows streaks and underestimates χ; closed-form L2 smooths; LSQR follows the data closely and keeps more noise and streaks.
- **Background removal** decides what counts as local. LBV keeps the tissue up to the mask's edge; V-SHARP is accurate inside but removes a rim at the edge; PDF can leave some background field near the edge. Errors there show as a smooth offset or shading in χ.
- **The mask** decides the reference (the mean over the mask) and the edge: a mask that includes noise or misses tissue changes all values.
- **Direction.** White matter's χ depends on its orientation to the main field; the map gives one value per voxel.
- **Fat** has a chemical shift that the field map takes for susceptibility. With water & fat ticked as well, the fat's phase is removed with the fat fraction of the water & fat fit before the field map is made, which needs echo times that suit water & fat (section 8); without it, fat in the body shows as false χ.
- **Short T_2^*** (dense iron, near air or bone) leaves little signal at the later echoes; the field there is noisy and χ unreliable.
- **Flow** in vessels and **motion** add phase that is not susceptibility.
- QSM needs a 3D scan with near-isotropic voxels (section 8).

T_1, inversion recovery. Each pixel is fitted to

> $Signal = | A - B \exp(-TI / T_1^*) |$, $T_1 = (B / A - 1) T_1^*$

T_1, saturation recovery. The signal as a function of TR is fitted to

> $Signal = A (1 - \exp(-TR / T_1))$

T_1, variable flip angle. The spoiled gradient echo signal over the flip angles is fitted per pixel with DESPOT1; a series acquired as a CINE is fitted per cardiac frame.

ORACLE. A balanced SSFP scan repeated with N RF phase increments gives every voxel a profile over the phase increments, whose shape depends on T_1, T_2, the flip angle and the off-resonance, and whose size on M_0. MAPapp calculates T_1, T_2 and the off-resonance from the profile directly, without fitting a

curve by trial; M_0 is the scale that matches the predicted profile to the measured one. R^2 compares the profile the four results predict with the measured profile, so min R-square works as for the other scans. The off-resonance is given in Hz, between −1/(2 TR) and +1/(2 TR).

Model-based fits. The maps are estimated from the k-space itself: BART looks for the maps whose predicted k-space matches the measured one.

8. Scan protocols for water & fat and QSM

Water & fat and QSM are calculated from multi-gradient echo scans. MAPapp offers them only when the scan allows it (Step 5). The tables below give starting points for mice at 3 T, 7 T and 9.4 T; adapt the field of view and resolution to the animal and the organ.

Water & fat

What matters most:

- **No fat suppression.** With the fat saturated there is no fat signal to separate.
- **Echo spacing.** Fat and water are back in phase after the *in-phase period*: 2.30 ms at 3 T, 0.99 ms at 7 T, 0.73 ms at 9.4 T. An echo spacing of a whole number of in-phase periods (or half of one) sees fat the same way at every echo, and the scan cannot be separated (MAPapp then switches **water & fat** off and says why); close to it, the separation is poor (MAPapp warns). The echo spacings in the tables lie about a third or two thirds of a period past a whole number of periods, which separates water and fat best. They are short in absolute terms but reachable with ordinary gradients; set them to within about 0.05 ms (at 9.4 T the whole period is only 0.73 ms).
- **Bandwidth.** Fat is shifted against water along the readout by (fat frequency) / (bandwidth per pixel) pixels; the fat frequency is 430 Hz at 3 T, 1010 Hz at 7 T and 1360 Hz at 9.4 T. Use the highest bandwidth the echo spacing allows: with the values in the tables the shift stays at about one to two pixels.
- **Flip angle.** Fat recovers faster than water (shorter T_1); a small flip angle (3D) or a long TR (2D) keeps the fat fraction from being overestimated.
- **Readout.** Monopolar (fly-back) readouts are best; bipolar readouts are corrected by MAPapp.

2D multi-slice multi-gradient echo

	3 T	7 T	9.4 T
First echo time	2.0–2.5 ms	2.0–2.5 ms	2.0–2.5 ms
Echo spacing	3.05 ms (or 3.8 ms)	2.65 ms (or 3.3 ms)	2.45 ms (or 3.2 ms)
Echoes	8 (last echo about 24 ms)	8 (about 21 ms)	8 (about 19 ms)
Bandwidth per pixel	≥ 300 Hz	≥ 500 Hz	≥ 600 Hz
TR	400–800 ms	400–800 ms	400–800 ms
Flip angle	20–30°	20–30°	20–30°
Resolution (mouse)	0.15–0.2 mm in plane, slices 0.5–1 mm	0.1–0.15 mm, slices 0.5–1 mm	0.1 mm, slices 0.5 mm
Averages	1–2	1–2	1–2
Fat suppression	off	off	off

A shorter TR (100–200 ms) gives more slices per minute; use a smaller flip angle with it (15–20°), so that the faster recovery of fat raises the fat fraction less.

3D multi-gradient echo

	3 T	7 T	9.4 T
First echo time	2.0–2.5 ms	2.0–2.5 ms	2.0–2.5 ms
Echo spacing	3.05 ms (or 3.8 ms)	2.65 ms (or 3.3 ms)	2.45 ms (or 3.2 ms)
Echoes	8 (last echo about 24 ms)	8 (about 21 ms)	8 (about 19 ms)
Bandwidth per pixel	≥ 300 Hz	≥ 500 Hz	≥ 600 Hz
TR	15–30 ms	15–30 ms	15–30 ms
Flip angle	3–6°	3–6°	3–6°
Resolution (mouse, isotropic)	0.25–0.3 mm	0.15–0.2 mm	0.12–0.15 mm
Averages	1	1	1
Fat suppression	off	off	off

Avoid the in-phase echo spacings: 2.30 or 4.60 ms at 3 T, multiples of 0.99 ms at 7 T (2.0, 3.0, 4.0 ms) and of 0.73 ms at 9.4 T (2.2, 2.9, 3.7 ms). Both readout types work at these spacings; the longer spacing in brackets leaves more time for a monopolar readout or a lower bandwidth.

QSM

QSM needs a **3D** multi-gradient echo scan: the susceptibility is calculated from the field in all three directions, which 2D slices do not give, so MAPapp does not offer QSM for 2D scans.

What matters most:

- **Isotropic voxels** and a field of view that covers the whole organ.
- **Echo times** up to about the T_2^* of the tissue: the phase is measured best there.
- **One average.** The scanner sums averages, and averages acquired at slightly different frequencies do not add up well; MAPapp checks and corrects this, but one average is safer.
- **Fully sampled k-space,** or compressed sensing (**CS reco** first; it takes longer).
- **The scan's orientation** is read from the data; keep the animal still and the field of view as planned.
- In the body, fat disturbs the phase: choose an echo spacing from the water & fat tables, so that the same scan also gives water and fat.

3D multi-gradient echo for QSM (mouse brain)

	3 T	7 T	9.4 T
Resolution (isotropic)	0.2–0.25 mm	0.12–0.15 mm	0.1 mm
First echo time	2.5–3 ms	2.5 ms	2.5 ms
Echo spacing	3.5–4 ms	3 ms	2.5–3 ms
Echoes	8 (last echo 25–30 ms)	8 (last echo 20–25 ms)	8 (last echo 18–20 ms)
TR	last echo + 5–10 ms	last echo + 5–10 ms	last echo + 5–10 ms
Flip angle	10–15°	10–15°	10–15°
Readout	monopolar, flow compensated if available	monopolar	monopolar
Averages	1	1	1

9. Credits

- **ORACLE**: N.M.J. Plähn, Y. Safarkhanlo, B.C. Açikgöz, A.L.C. Mackowiak, P. Radojewski, G. Bonanno, E.S. Peper, R. Wiest, J.A.M. Bastiaansen, ORACLE: an analytical approach for T1, T2, proton density, and off-resonance mapping with phase-cycled balanced steady-state free precession, Magn Reson Med 93(4), 1657–1673 (2025).
- **BART toolbox**: mrirecon.github.io/bart. M. Uecker et al., Berkeley Advanced Reconstruction Toolbox, Proc. Intl. Soc. Mag. Reson. Med. 23, 2486 (2015).
- **sigpy**: Frank Ong and Michael Lustig, SigPy: a python package for high performance iterative reconstruction, Proc. Intl. Soc. Mag. Reson. Med. 27, 4819 (2019).
- **elastix**: S. Klein, M. Staring, K. Murphy, M.A. Viergever, J.P.W. Pluim, elastix: a toolbox for intensity based medical image registration, IEEE Transactions on Medical Imaging 29(1), 196–205 (2010). elastix.dev
- **PCA denoising**: J. Veraart, D. S. Novikov, D. Christiaens, B. Ades-Aron, J. Sijbers, E. Fieremans, Denoising of diffusion MRI using random matrix theory, NeuroImage 142, 394–406 (2016).
- **Gibbs ringing suppression**: E. Kellner, B. Dhital, V.G. Kiselev, M. Reisert, Gibbs-ringing artifact removal based on local subvoxel-shifts, Magn Reson Med 76(5), 1574–1581 (2016).
- **Water & fat separation**: the ISMRM Fat-Water Toolbox: H.H. Hu, P. Börnert, D. Hernando et al., ISMRM workshop on fat-water separation: insights, applications and progress in MRI, Magn Reson Med 68(2), 378–388 (2012); its fw3pluspoint method: S.D. Sharma, H.H. Hu, K.S. Nayak, Accelerated water-fat imaging using restricted subspace field map estimation and compressed sensing, Magn Reson Med 67(3), 650–659 (2012).
- **DESPOT1**: S.C.L. Deoni, B.K. Rutt, T.M. Peters, Rapid combined T1 and T2 mapping using gradient recalled acquisition in the steady state, Magn Reson Med 49(3), 515–526 (2003).
- **Water & fat separation methods**: graph cut: D. Hernando, P. Kellman, J.P. Haldar, Z.-P. Liang, Robust water/fat separation in the presence of large field inhomogeneities using a graph cut algorithm, Magn Reson Med 63(1), 79–90 (2010), with V. Kolmogorov, R. Zabih, What energy functions can be minimized via graph cuts?, IEEE Trans Pattern Anal Mach Intell 26(2), 147–159 (2004); hierarchical IDEAL: J. Tsao, Y. Jiang, Hierarchical IDEAL: fast, robust, and multiresolution separation of multiple chemical species from multiple echo times, Magn Reson Med 70(1), 155–159 (2013); region growing: H. Yu, S.B. Reeder, A. Shimakawa, J.H. Brittain, N.J. Pelc, Field map estimation with a region growing scheme for iterative 3-point water-fat decomposition, Magn Reson Med 54(4), 1032–1039 (2005); IDEAL: S.B. Reeder, A.R. Pineda, Z. Wen et al., Iterative decomposition of water and fat with echo asymmetry and least-squares estimation (IDEAL), Magn Reson Med 51(1), 35–45 (2004); H. Yu, C.A. McKenzie, A. Shimakawa et al., Multiecho reconstruction for simultaneous water-fat decomposition and T2* estimation, J Magn Reson Imaging 26(4), 1153–1161 (2007); variable projection: G.H. Golub, V. Pereyra, The differentiation of pseudo-inverses and nonlinear least squares problems whose variables separate, SIAM J Numer Anal 10(2), 413–432 (1973); D. Hernando, J.P. Haldar, B.P. Sutton, J. Ma, P. Kellman, Z.-P. Liang, Joint estimation of water/fat images and field inhomogeneity map, Magn Reson Med 59(3), 571–580 (2008); the fat fraction: H. Yu, A. Shimakawa, C.D.G. Hines et al., Combination of complex-based and magnitude-based multiecho water-fat separation for accurate quantification of fat-fraction, Magn Reson Med 66(1), 199–206 (2011); D. Hernando, C.D.G. Hines, H. Yu, S.B. Reeder, Addressing phase errors in fat-water imaging using a mixed magnitude/complex fitting method, Magn Reson Med 67(3), 638–644 (2012); the bipolar readout: W. Lu, H. Yu, A.

Shimakawa, M. Alley, S.B. Reeder, B.A. Hargreaves, Water-fat separation with bipolar multiecho sequences, Magn Reson Med 60(1), 198–209 (2008); coil combination: D.O. Walsh, A.F. Gmitro, M.W. Marcellin, Adaptive reconstruction of phased array MR imagery, Magn Reson Med 43(5), 682–690 (2000).

- **Susceptibility mapping (QSM)**: the dipole field: R. Salomir, B.D. de Senneville, C.T.W. Moonen, A fast calculation method for magnetic field inhomogeneity due to an arbitrary distribution of bulk susceptibility, Concepts Magn Reson B 19(1), 26–34 (2003); J.P. Marques, R. Bowtell, Application of a Fourier-based method for rapid calculation of field inhomogeneity due to spatial variation of magnetic susceptibility, Concepts Magn Reson B 25(1), 65–78 (2005). Phase unwrapping: W. Xu, I. Cumming, A region-growing algorithm for InSAR phase unwrapping, IEEE Trans Geosci Remote Sens 37(1), 124–134 (1999). Background field removal: LBV, D. Zhou, T. Liu, P. Spincemaille, Y. Wang, Background field removal by solving the Laplacian boundary value problem, NMR Biomed 27(3), 312–319 (2014); V-SHARP, F. Schweser, A. Deistung, B.W. Lehr, J.R. Reichenbach, Quantitative imaging of intrinsic magnetic tissue properties using MRI signal phase, NeuroImage 54(4), 2789–2807 (2011), and B. Wu, W. Li, A. Guidon, C. Liu, Whole brain susceptibility mapping using compressed sensing, Magn Reson Med 67(1), 137–147 (2012); PDF, T. Liu, I. Khalidov, L. de Rochefort et al., A novel background field removal method for MRI using projection onto dipole fields (PDF), NMR Biomed 24(9), 1129–1136 (2011). Dipole inversion: TKD, K. Shmueli, J.A. de Zwart, P. van Gelderen et al., Magnetic susceptibility mapping of brain tissue in vivo using MRI phase data, Magn Reson Med 62(6), 1510–1522 (2009); LSQR, C.C. Paige, M.A. Saunders, LSQR: an algorithm for sparse linear equations and sparse least squares, ACM Trans Math Softw 8(1), 43–71 (1982); TV and closed-form L2, B. Bilgic, A.P. Fan, J.R. Polimeni et al., Fast quantitative susceptibility mapping with L1-regularization and automatic parameter selection, Magn Reson Med 72(5), 1444–1459 (2014); C. Milovic, B. Bilgic, B. Zhao, J. Acosta-Cabronero, C. Tejos, Fast nonlinear susceptibility inversion with variational regularization, Magn Reson Med 80(2), 814–821 (2018).
- **Otsu's threshold**: N. Otsu, A threshold selection method from gray-level histograms, IEEE Trans Syst Man Cybern 9(1), 62–66 (1979).
- **Color maps**: F. Crameri, Scientific colour maps (navia, lipari), Zenodo, doi:10.5281/zenodo.1243862.
- **NIfTI conversion**: dcm2niix: C. Rorden et al., github.com/rordenlab/dcm2niix.